

Lab 5: Viewing Life — Observing Structure and Function

BIOL-1

Name: _____ **Date:** _____

Overview

Biology is the study of life. In this lab, you will move beyond the textbook and observe living systems directly. Using a compound light microscope, you will examine a variety of specimens—from plant cells to pond water organisms—to identify common features of life and appreciate its diversity. This lab focuses on **observation**, **scientific illustration**, and **connecting structure to function**.

Learning Objectives

By the end of this lab, you will be able to:

1. Use a compound microscope to locate and focus on living specimens.
2. Create accurate scientific drawings of biological specimens.
3. Identify common structures (cell walls, chloroplasts, cytoplasm) in different organisms.
4. Relate observed structures to their functions (e.g., green chloroplasts = photosynthesis).
5. Appreciate the complexity and diversity of life at the microscopic scale.

Materials

- Compound light microscope
- Slides and coverslips
- *Elodea* (freshwater plant) leaves
- Pond water samples (containing various protists/algae)
- Droppers/pipettes
- Lens paper

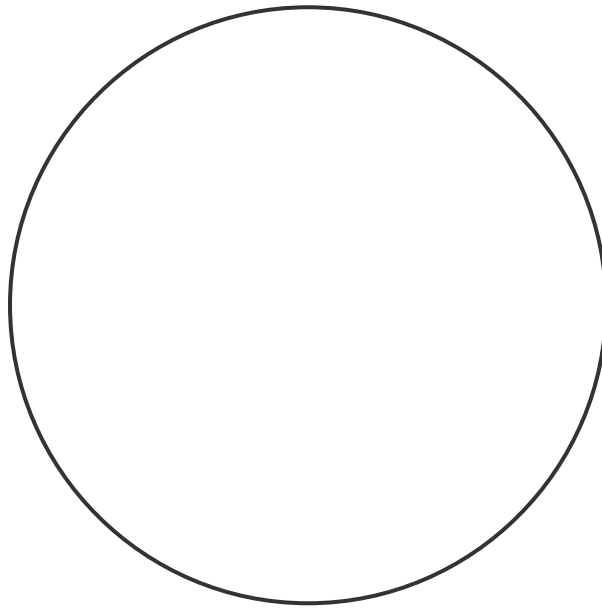
Part 1: The Plant Cell — *Elodea*

Elodea is a common aquatic plant. Its leaves are only a few cells thick, making them perfect for viewing under a microscope.

Procedure:

1. Place a single *Elodea* leaf on a slide.
2. Add a drop of water and a coverslip.
3. Focus under **Low Power (4×)**, then move to **Medium (10×)** and **High Power (40×)**.

Draw a single *Elodea* cell at High Power (400× total magnification). Label the CELL WALL and CHLOROPLASTS.



Magnification used:

1. What shape are the cells? How does the cell wall contribute to this shape?

2. Describe the movement of the chloroplasts (cytoplasmic streaming), if observed. Why might they move?

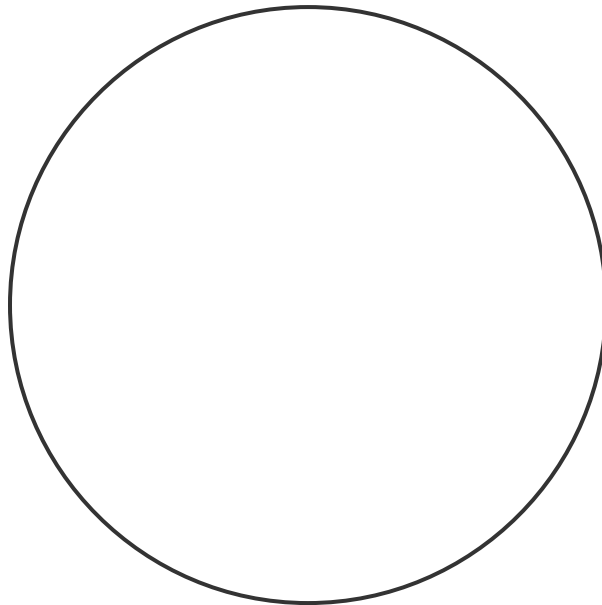
Part 2: The Microscopic World — Pond Water

Pond water is a teeming ecosystem. You may see algae (green, non-moving or moving), protozoa (swimming, crawling), or small animals (rotifers, crustaceans).

Procedure:

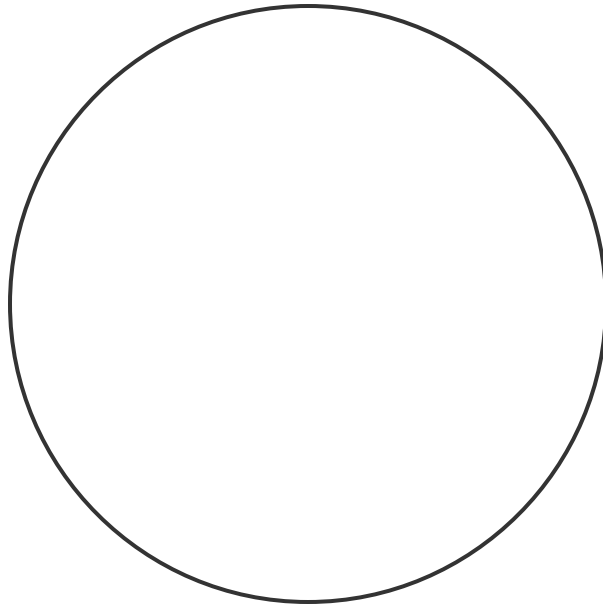
1. Place a drop of pond water on a clean slide.
2. Add a coverslip carefully.
3. Scan the slide under **Low Power (4×)** to find movement or interesting objects.
4. Switch to higher power to observe details.

Draw the first organism you find. Is it green? Does it move?



Magnification used:

Draw a second, different organism.



Magnification used:

1. Compare the two organisms you drew. How are they similar? How are they different?

2. Did you observe any interaction between organisms? (e.g., predation, avoiding each other)

Part 3: Synthesis — The Unity and Diversity of Life

All the things you viewed today are **alive**. They all reproduce, use energy, and are made of cells.

1. Based on your observations today, what is one feature that ALL these specimens seemed to have in common?

2. How does the structure of the *Elodea* cell (rigid, green) relate to its lifestyle (stationary, photosynthetic)?

3. How does the structure of the pond water organisms (cilia, flagella, flexible bodies) relate to their lifestyle?

Conclusion

Life is diverse in form but unified in fundamental structure. By observing these organisms, we connect the abstract concepts of biology (cells, energy, evolution) to the reality of the living world.